

Applying testing techniques

There are several testing techniques which may be used as per the requirement of the game and designers. Trying to test every possible combination of game event, function, and configuration is neither practical nor economical. One popular technique is combinatorial testing.

- ▶ **Combinatorial Testing:** Here you choose parameters like game events, gameplay options, hardware configurations, character choices etc and test them pair wise. Here combinatorial tables are constructed with parameters to be tested. For example whether a player chooses to play as Mario or Luigi, chooses a black or a white suit etc.
- ▶ **Test Flow Diagrams:** One other technique is to use test flow diagrams. Test flow diagrams are a formal approach to testing with a high degree of reusability in sequels, similar game events etc. They are easy to analyze and understand because of their graphical nature and it is easy to take feedback from developers.
- ▶ **Test Trees:** Test trees are a set of tree graphs made to indicate the testing process. They are used to document and understand hierarchical association of test cases and game features and functions. They can be made using pen and paper or using some software. Every time a new build is sent for testing, these test trees are updated.

Bug fixing and verification

After the developers fix the bug as directed by the game designer or technical director, this bug fix has to be verified by the testers. Once verified, the bug is deleted from the bug list. However some bugs need not be fixed as they are minor and thus are waived by game designers and producers.

- ▶ **Quality Assurance:** This process takes place through-out the Production phase. Quality Assurance is the systematic monitoring and evaluation of various aspects of the game. It focuses on the quality of the game and aims at assuring best possible quality. The most important method for successful QA is a formal review system that covers all aspects of Design and Production. The following are the testing methods followed in QA:

Final testing

This follows the traditional software testing mechanism of Software Engineering. Here, the code of the game is tested by conducting a series of Formal and Informal tests, which are already discussed.